

CONTACT INFORMATION	AI Center, ETH Zürich X22.1 OAT Andreasstrasse 5, 8050 Zürich, Switzerland	<i>e-mail:</i> ischlag@ethz.ch
RESEARCH INTERESTS	My research focuses on developing transparent and responsible AI models, in particular large language models, so that Switzerland and Europe retain the capability to develop trustworthy, fully transparent AI as a public good, independent of closed commercial alternatives. I am the ETH Zurich lead of the Apertus project, leading a team of 12+ full-time engineers and a larger number of academic contributors (MSc and PhD students, postdocs) on the development of Apertus on Switzerland's Alps supercomputer at CSCS. Apertus 70B is one of the largest fully-open LLMs developed by a public institution, with open weights, open data, and in particular open values set out in the Apertus charter, and among the first at this scale to be developed responsibly: fully documented training data, broad multilingual coverage, and respect for data owners' opt-outs. Training scaled to 4,096 NVIDIA Grace Hopper GPUs and consumed several million GPU-hours. Apertus 1.5 extends this with an efficient and scalable approach to multimodal training, reaching performance competitive with Gemma 3 and Gemma 4, stronger instruction following, a reinforcement learning phase for code and mathematics, and a longer context window, improving over Apertus 1.0 across all benchmarks. My earlier work on Fast Weight Programmers during my PhD at USI laid the foundation for my current work on Transformer attention alternatives such as DeltaNet, now deployed in production models such as the Qwen family since Qwen 3.5, Kimi Linear and Kimi K3 (at 2.8T parameters one of the largest open-weight models to date), and OLMo Hybrid. As a lecturer at ETH Zurich, I teach Large-Scale AI Engineering, training the next generation of researchers and engineers to develop and operate large models on modern supercomputing infrastructure.	
CURRENT	AI Research Scientist, AI Center, ETH Zürich	since 02/2024
	• Co-leading the Apertus project and its core engineering team • Released Apertus 8B and 70B (09/2025) and Apertus 1.5 (07/2026), fully-open, responsibly developed LLMs trained on Alps at CSCS • Teaching the MSc course <i>Large-Scale AI Engineering</i>	
EDUCATION	Università della Svizzera italiana, Lugano, Switzerland PhD, Artificial Intelligence and Machine Learning, March 2023 • With distinction • Thesis: <i>Fast Weight Programmers for Greater Systematic Generalisation in Language</i> • Adviser: Prof. Dr. Jürgen Schmidhuber University of St Andrews, St Andrews, Scotland MSc, Artificial Intelligence, August 2016 • With distinction • Thesis: <i>Face Recognition from Ancient Roman Coins</i> • Adviser: Prof. Dr. Ognjen Arandjelović University of Applied Sciences and Arts Northwestern Switzerland, Brugg, Switzerland BSc, Computer Science, August 2015 • With specialisation in <i>Information Processing and Visualization</i> • Thesis: <i>Face Similarity - Finding Lookalikes from Images</i> • Adviser: Prof. Dr. Manfred Vogel	

TEACHING EXPERIENCE	Swiss Armed Forces Special Forces Training Center , Isonne, Switzerland	
	• Basic Training, 2010 • Non-commissioned Officer School, 2011	
	ETH Zürich , Switzerland	since 2023
	<i>Lecturer</i> • Large-Scale AI Engineering every semester since 02/2025 (80, 125 and 160 participants in the first three editions) • AI Center Projects in Machine Learning Research every semester since 02/2025	
	<i>Head teaching assistant</i> • Deep Learning Fall 23/24	
	<i>Guest lectures</i> • EPFL, CS-461 Machine Learning for Natural Language Processing, 10/2025 • University of Lucerne, Ethics of Digital Transformation and AI, 11/2025	
	Università della Svizzera italiana , Lugano, Switzerland	2017 - 2021
	<i>Teaching assistant</i> • Machine Learning Fall 17/18 • Deep Learning Lab Fall 18/19, Fall 19/20 • Graph Deep Learning Spring 20/21	
	<i>Course development</i> • Assisted the development of the first version of the Deep Learning Lab for Fall 18/19	
	Swiss Armed Forces, KSK, Gren Bat 30/2 , Isonne, Switzerland	2012 - 2019
PROFESSIONAL EXPERIENCE	<i>Military instructor and squad leader in the Special Forces Command (KSK)</i> • A yearly 4 week repetition course	
	ETH Zürich , Zürich, Switzerland	08/2023 - 01/2024
	<i>Postdoctoral Researcher under Prof. Dr. Thomas Hofmann</i>	
	Università della Svizzera italiana , Lugano, Switzerland	05/2023 - 7/2023
	<i>Postdoctoral Researcher under Prof. Dr. Jürgen Schmidhuber</i>	
	Università della Svizzera italiana , Lugano, Switzerland	09/2016 - 4/2023
	<i>Doctoral Researcher under Prof. Dr. Jürgen Schmidhuber</i>	
	Meta AI , Menlo Park, California, USA	07/2022 - 10/2022
	<i>Research internship at FAIR with Dr. Xian Li and Dr. Jason Weston</i>	
	Google Research , Mountain View, California, USA	09/2020 - 02/2021
	<i>Remote research internship in the Blueshift Team with Dr. Behnam Neyshabur</i>	
	Microsoft Research , Redmond, Washington, USA	06/2019 - 09/2019
	<i>Research internship with Prof. Dr. Paul Smolensky</i>	

Basler Kantonalbank, Basel, Switzerland
Computer science apprenticeship

09/2006 - 06/2010

AWARDS

Global Swiss AI Award 2025, Special Recognition

Mindfire Foundation, for the development of Apertus. Received January 2026.

Culture & Society "AI Award Night"

For the development of Apertus. Received 2025.

NVAIL Pioneering Research Award

For *Learning to Reason with Third-Order Tensor Products*. Received at NeurIPS, 2018.

University of St Andrews

Medal for the best dissertation in Computer Science, 2016

GRANTS

A Swiss-Centric Foundation Model for Switzerland's Sovereign AI Future 2025
Democratizing LLMs for Global Languages with Mixtures of Multilingual Experts 2025

ADVISORY

EinstAI
Swiss AI Association (formerly KImpact)
vrenni.ai, Apertus-based AI for SMEs

REVIEWING

Area chair: NeurIPS 2026. Reviewer: NeurIPS 19/20/21, ICML 20/21/25, ICLR 20/21/22